



Society of Petroleum Engineers

SPE-227138-MS

A New Practical Anhydrite Classification Scheme: Implications for Depositional Environments and Seal Assessment

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227138-MS](https://doi.org/10.2118/227138-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The absence of a comprehensive classification system for anhydrites has led to oversimplified assumptions of homogeneity, which undermines accurate seal evaluations. This study presents a new classification scheme that captures the heterogeneity of anhydrite deposits and enhances interpretations of depositional environments and seal integrity. A 70-meter Late Jurassic core (Arab A, B, and C members) near Riyadh, Saudi Arabia, was analyzed using sedimentological and mineralogical techniques, including over 120 thin sections, high-resolution gamma-ray readings, and porosity measurements. Four structural anhydrite groups with distinct gamma-ray signatures were identified and categorized by their anhydrite-to-carbonate ratios into eleven classes. These classes reflect varying depositional and diagenetic settings with distinct petrophysical properties. Additionally, a novel anhydrite logging sheet is introduced to aid in core and outcrop analysis, improve facies recognition, and predict gamma-ray responses. The study emphasizes the importance of detailed anhydrite characterization for reliable subsurface seal assessments.

Introduction

Seal integrity is vital for hydrocarbon entrapment but is often overlooked, despite evidence that over half of dry wells result from inadequate seals (Karlo, 2018; Rudolph & Goulding, 2017). The Arab-Hith evaporites are key caprocks in major petroleum systems (Ye et al., 2023), yet seal failures in Qatar highlight gaps in understanding their behavior (Van Buchem et al., 2014). A major challenge is the lack of a standardized classification system for evaporites, unlike those used for carbonates and clastics (Boggs, 2009). Anhydrite's post-depositional nature and frequent recrystallization complicate environmental interpretations; however, preserved gypsum structures can offer valuable insights (Babel & Schreiber, 2014; Kasprzyk, 2003). This study analyzes a Late Jurassic Arab-C, B, A core, introduces a new anhydrite classification scheme, and evaluates its application to sedimentological interpretation and seal assessment.

Data and Method

This study analyzes a 70-meter shallow core encompassing the Late Jurassic Arab A, B, and C members. The core, acquired by King Abdullah University of Science and Technology near Riyadh, Saudi Arabia,

underwent detailed sedimentological and mineralogical characterization using over 120 petrographic thin sections. The primary objective of the core description is to document the heterogeneity of both carbonate and evaporite facies. To evaluate whether lithofacies variability correlates with distinct well-log signatures, high-resolution core gamma-ray measurements were collected at 30-cm intervals and complemented by wireline gamma-ray logs from the same well. Porosity was assessed through a log constructed from plug measurements obtained using a Helium Porosimeter, with sampling intervals of 60 cm in homogeneous sections and 30 cm in heterogeneous intervals. Porosity analysis is aimed at evaluating the integrity and seal potential of the various anhydrite facies.

Previous Anhydrite Classifications

[Maiklem et al. \(1969\)](#) were among the first to propose a systematic classification of anhydrite, introducing a unified terminology to describe its structural and textural variability. Despite its foundational nature, this scheme has seen limited application in subsequent anhydrite research ([Forkner, 2010](#)). Although the classification offers a comprehensive framework, it is not conducive to interpreting depositional environments, lacks concise descriptors for distinct structural forms, and comprises 25 structural and 6 textural categories, making it overly complex and impractical for routine use. [Meyer \(2005\)](#) proposed a classification system grounded in structural variation, crystal orientation, and aggregate morphology. Unlike the framework by [Maiklem et al. \(1969\)](#), Meyer's system integrates both microscopic and macroscopic features within a unified scheme. Developed as an extension of a calcium sulfate diagenesis study initiated in 2001, this classification remains under evaluation. [Forkner \(2010\)](#) proposed a simplified classification system targeting macro-scale anhydrite fabrics, departing from earlier schemes by assigning a comprehensive nomenclature to each fabric type. Emphasizing total anhydrite volume, morphology, and host sediment characteristics, the system aims to enhance the interpretation of original depositional environments.

Each previous classification addressed specific challenges in describing anhydrite, resulting in some overlap but also notable divergences. However, none adequately accounted for anhydrite's inherent metastability, which drives frequent recrystallization and structural-textural transformations ([Babel & Schreiber, 2014](#)). Additionally, the lack of a dedicated methodology and standardized logging sheet for anhydrite remains a significant gap ([Forkner, 2010](#)). This has resulted in inconsistent nomenclature in the literature, leading to miscommunication and misinterpretation, ultimately undermining the goal of standardizing terminology as initially envisioned by [Maiklem et al. \(1969\)](#).

Results

Facies and Petrophysical Analysis

The core is subdivided into three thick evaporite intervals, separated by two relatively thinner carbonate sections. A total of fifteen distinct lithofacies have been identified, including five carbonate and ten evaporite facies. Facies interpretation has been integrated into a composite log, incorporating core gamma-ray (GR) measurements, wireline GR data, plug-derived porosity values, and X-ray fluorescence (XRF) geochemical analysis. Anhydrite-rich intervals are predominantly associated with gamma-ray values below 23 API, whereas carbonate-rich sections are generally observed at values exceeding this threshold. All evaporite facies exhibit elevated concentrations of sulfur and calcium. In contrast, concentrations of light elements are comparatively lower within evaporitic intervals relative to the carbonate-dominated sections.

New Practical Anhydrite Classification Scheme

The proposed anhydrite classification targets bedded anhydrite—anhydrite deposits that form discrete sedimentary rock units, as defined by [Murray \(1964\)](#). Consequently, the other two modes of formation

—pore-filling and replacement anhydrite—do not comprise distinct sedimentary beds; rather, anhydrites are observed either as cement-filling voids between foreign sediments or as isolated crystals replacing pre-existing minerals (Murray, 1964; Warren, 2016). The second cornerstone of this classification acknowledges anhydrite's nature by evaluating the likelihood of original gypsum structure preservation during anhydratization. The third cornerstone centers on the structural variability of original gypsum, governed by factors including water depth, hydrodynamics, evaporation intensity, geochemical conditions, and salinity (Babel & Schreiber, 2014; Murray, 1964; Warren, 2016). Two main categories have been established based on the preservation of Gypsum's original structure. Each structure group classified in either category is further subdivided into four main classes based on the percentage of anhydrite or gypsum within the rock relative to other existing minerals or sediments: (1) Low Anhydrite (10-40%). (2) Medium Anhydrite (40-60%). (3) High Anhydrite (60-99%). (4) Very High Anhydrite (99-100%). The percentages have been estimated based on observations from the Arab anhydrite core. The rate of anhydrite in the rock indicates salinity, restriction conditions, aridity, proximity to brine, or intensity of burial diagenesis (Babel & Schreiber, 2014; Hardie L. A., 2003; Warren, 2016). In total, 11 classes are recognized and can be used to describe any anhydrite rock (Fig. 1).

PRACTICAL ANHYDRITE CLASSIFICATION SCHEME

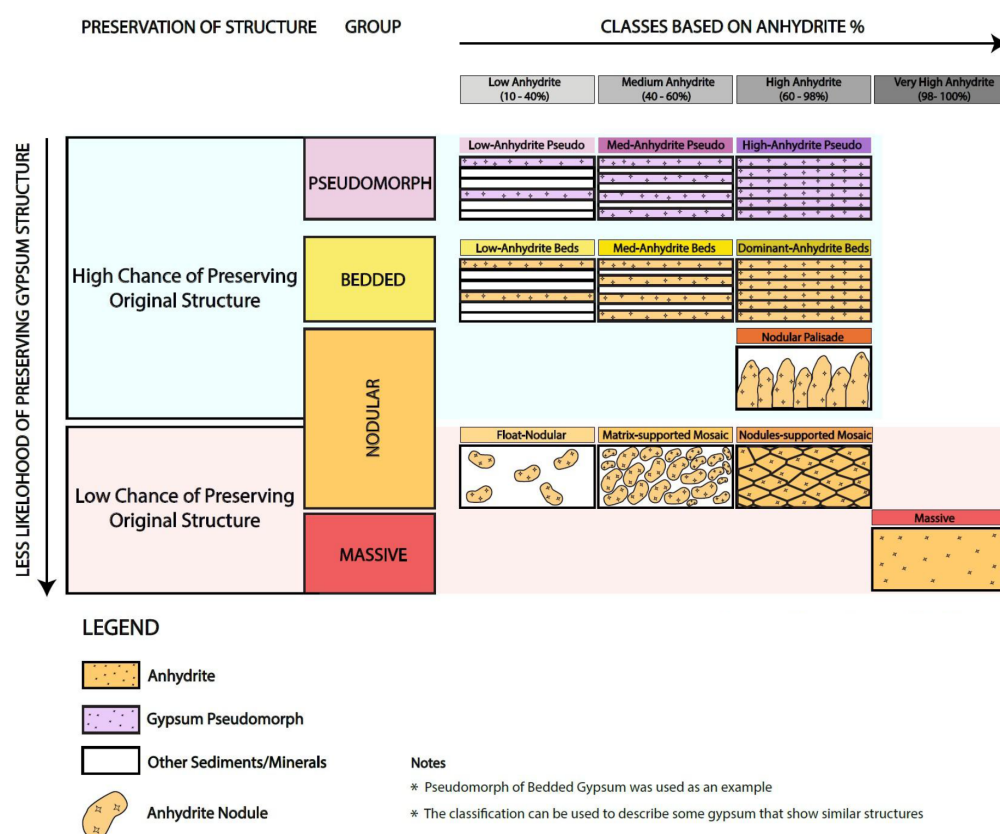


Figure 1—A new anhydrite classification scheme based on two factors: the likelihood of preserving the original gypsum structure (vertical axis) and the percentage of anhydrite relative to other minerals in the rock (horizontal axis). There are 11 classes to potentially describe any anhydrite rock.

High Chance of Preserving Original Structure. Structures that preserve their primary gypsum fabric, including:

1. Pseudomorphed Gypsum Group: Anhydrites that retain the morphology of the original gypsum replacement form a diverse group, reflecting various types of gypsum. These pseudomorphs are well

documented in the literature (Kerr Jr. & Thomson Alan, 1963; Maiklem et al., 1969; Meyer, 2005) and commonly exhibit rosette-like, pyramidal, or discoidal crystal habits (Maiklem et al., 1969; Meyer, 2005). This group offers the clearest insight into the origin and depositional environment of the anhydrite.

2. **Bedded Anhydrite Group:** This structure, typically ranging from millimeters to centimeters in thickness, is also referred to as laminated anhydrite (Fallatah et al., 2023; Forkner, 2010; Wolpert et al., 2024). It most likely originates from layered gypsum formations or reconstituted bottom-nucleated gypsum crystals (Kerr Jr. & Thomson Alan, 1963; Kirkland, 2003). The patterns and percentage of anhydrites can indicate brine hydrology, climate changes, and salinity (Kirkland, 2003; Warren, 2016).
3. **Nodular Palisade Class of the Nodular Group:** Vertically oriented nodular anhydrite represents a distinct subtype within the nodular anhydrite group, characterized by a unique origin (Warren, 2016). Unlike typical nodular forms, its vertical alignment facilitates the identification of its primary structure. As such, it is classified as having a high potential to retain original bottom-nucleated, vertically aligned gypsum crystals.

Low Chance of Preserving Original Structure. A category encompassing anhydrite structures in which the original gypsum fabric has been obliterated or overprinted by secondary structures, or is absent, including:

1. **Nodular Anhydrite Group:** Nodular anhydrite consists of clustered anhydrite masses, typically formed from gypsum crystals that develop during the post-depositional stage (Murray, 1964). However, Various gypsum types—such as meshwork gypsum, bottom-nucleated crystals, and reworked gypsolites—can also transform into nodular anhydrite (Warren, 2016).
2. **Massive Anhydrite Group:** Massive anhydrite refers to a structureless form of anhydrite lacking internal fabric or distinguishing features. It may form from any gypsum or anhydrite type during deposition, post-deposition, or through extensive burial diagenesis. Consequently, it represents the most interpretively challenging anhydrites (Kendall, 2005).

Anhydrite Facies Modifiers

Once a class is identified, additional features—such as nodule size, degree of distortion, and bedding patterns—can be used to define specific facies (Fig. 2). These modifiers aid in reconstructing the depositional and diagenetic history and inferring the original gypsum structures.

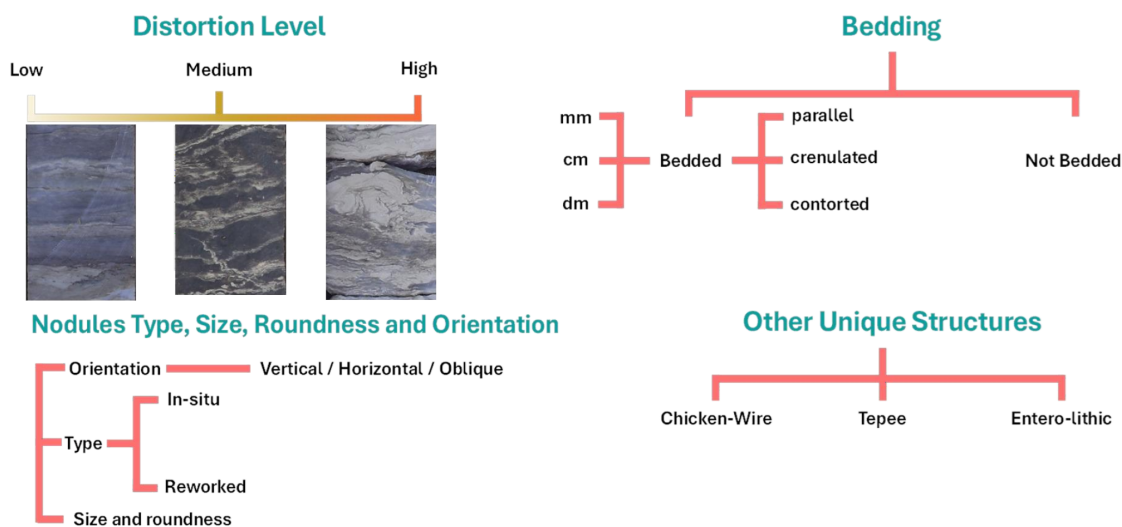


Figure 2—Facies modifiers for anhydrite can help differentiate between two facies of the same class, aiding interpretations.

Anhydrite Specialized Logging Template (Fig. 3)

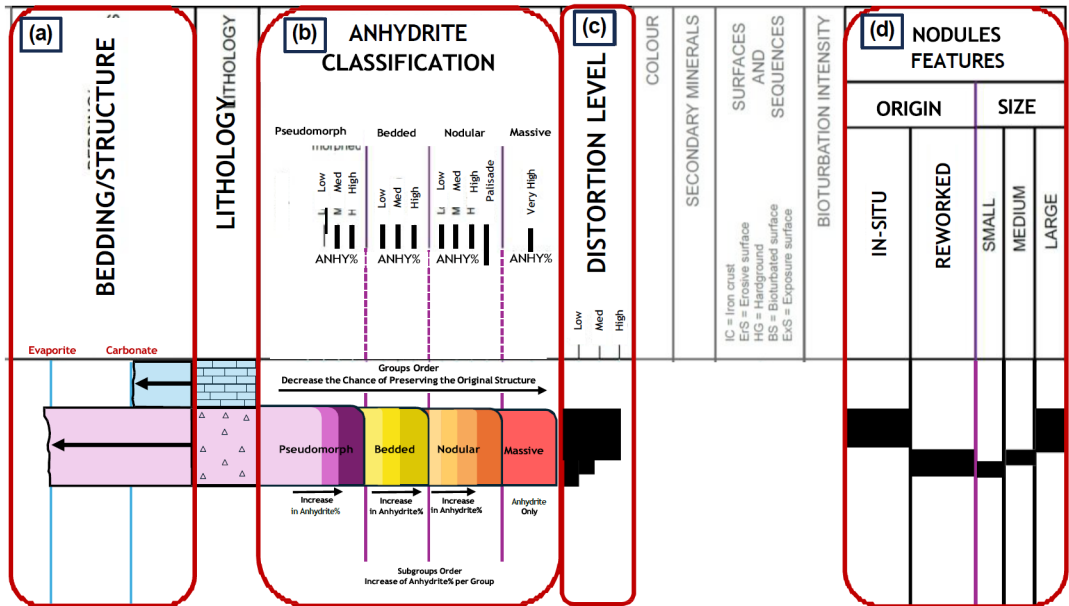


Figure 3—A practical anhydrite logging sheet designed for the new classification scheme. (a) Evaporite bedding is more prominent, depicting low GR signatures. (b) Anhydrite groups are arranged from left to right based on preservation potential; within each group, anhydrite% increases to the right. (c) distortion level. (d) Dedicated for the nodular group, nodules size and origin can be indicated.

Anhydrite Classification Column. The new Anhydrite Classification scheme is integrated into the texture description column by arranging the four main groups— Pseudomorph, Bedded, Nodular, and Massive— from left to right, indicating decreasing potential for preserving original gypsum structures. Within each group, three subgroups represent increasing anhydrite content, reflecting progressive evaporation, rising salinity, or changes in water chemistry. To improve visual clarity, colored vertical lines separate the groups across the logging sheet.

Bedding/Sedimentary Structures Column. A revised format of the traditional column is presented to depict evaporite units. Two vertical parallel lines have been added to indicate the extent of each lithological unit when plotted from right to left. Evaporite units extend farther than carbonate units, reflecting their typically lower gamma-ray signatures. Anhydrite sections were characterized by gamma rays below 23 API. Applying this method to carbonate–evaporite successions facilitates the prediction of gamma-ray trends within the described interval. The methodology was evaluated by comparing the described core against both wireline and core-acquired gamma-ray data of the section.

Distortion Intensity. Indicating the degree of distortion in anhydrite facies helps identify intervals with significant diagenetic overprints and supports the evaluation of distortion patterns across various facies.

Nodules Origin and Size. Indicating whether nodules are in situ or mechanically reworked and recording their relative sizes in nodular anhydrite facies and isolated nodules within other facies.

Discussion

Implications of the New Classification on Anhydrite Description and Its Broader Basins Applications

The newly developed method allows for more precise documentation of gradual lithological transitions between carbonate and evaporite facies, which is crucial for interpreting environmental settings and

petrophysical properties (Fig. 3) (Kirkland, 2003; Sarg, 2001). In comparison, previous research often portrayed these transitions as sharp boundaries, thereby oversimplifying the complexities of the depositional record (Fig. 4). Additionally, conventional logging sheets frequently misrepresent variations in anhydrite structures, stacking patterns, and facies boundaries, typically as straight vertical lines.

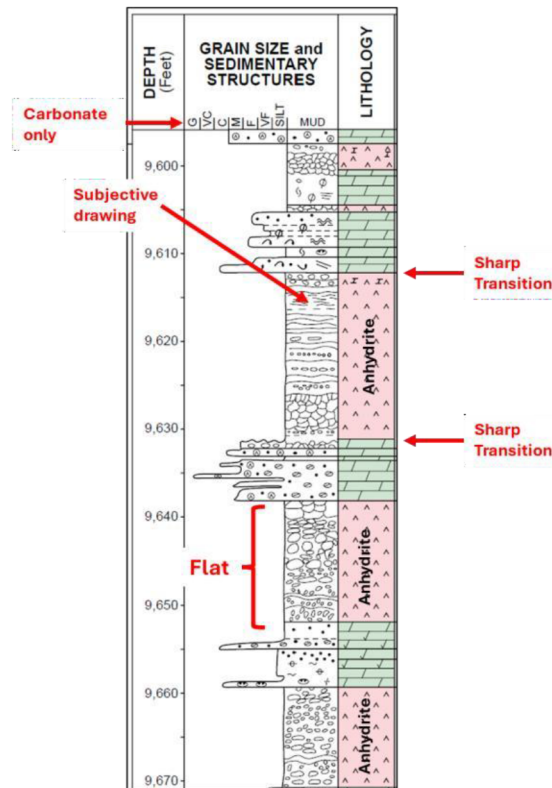


Figure 4—Conventional Anhydrite Logging Profile (Azer & Peebles, 1998). This profile illustrates a typical method for anhydrite description, which often generates unrealistically sharp contacts between anhydrite and surrounding carbonate or clastic lithologies. A significant limitation of this approach is its tendency to represent all anhydrite structures as a featureless flat line, failing to capture inherent textural variations. Furthermore, detailed textural depiction relies heavily on subjective graphical representations.

The classified anhydrite facies in this study have been recognized globally across various evaporite basins, indicating the potential applicability of this facies classification system in diverse geological settings. For example, bedded anhydrite with varying carbonate- to-anhydrite ratios has been documented by Kirkland (2003) in the Castile Formation varves of Texas. Nodular anhydrite facies have been widely reported in the Oligocene to Early Miocene Asmari Formation (Omidpour et al., 2023), as well as in the Badenian evaporite deposits of Poland (Kasprzyk, 2003). Furthermore, Wolpert et al. (2024), in their sedimentological analysis of the Late Jurassic Hith Formation outcrops in Saudi Arabia, identified two distinct types of nodular anhydrite, differentiated by matrix content. Pseudomorphed gypsum has been described by Maiklem et al. (1969) in the Devonian anhydrites of the Elk Point Basin in Alberta. Massive, structureless anhydrite facies have also been reported by Maiklem et al. (1969) and in the Canning Basin evaporites of Western Australia (Warren & Kempton, 1997).

Implications of the New Anhydrite Classification Scheme on Depositional Environment

The proposed classification scheme provides a systematic framework for interpreting anhydrite facies and consisted of the following steps: (1) classifying the anhydrite facies; (2) identifying key facies modifiers; (3) inferring the original gypsum structure; (4) determining whether the origin is depositional, post-depositional, or diagenetic; (5) defining the stacking pattern, thickness, and bounding surfaces within the

evaporite–carbonate succession; and (6) interpreting the depositional environment. By emphasizing the concept of preservation potential for original gypsum structures and anhydrite content, the classification scheme differentiates distinct anhydrite facies through analysis of the host carbonate matrix, highlighting the need for careful examination of anhydrite.

Implications of the New Anhydrite Classification Scheme on Seal Assessment

Porosity data from the Arab core show that carbonate facies generally exhibit higher porosity than anhydrite facies. However, anhydrite facies with less than 40% anhydrite content displayed some porosity largely due to the influence of the more porous carbonate host matrix. Recent work on anhydrite seals highlighted the potential impact of anhydrite structures on seal strength, demonstrating that both composition and structural type could affect seal integrity (Hernandez, 2024). As such, the classification scheme could potentially improve the resolution and applicability of anhydrite seal analysis in both modeling and field-based assessments, utilizing three-dimensional modeling based on detailed sequence stratigraphic frameworks, similar to those employed in reservoir characterization (Karlo, 2018; Simmons, n.d.).

Conclusion

This study introduced a new anhydrite classification method and description methodology. Using a custom logging sheet, ten distinct anhydrite classes were identified. Petrophysical data showed that certain facies have porosities that may compromise seal integrity due to the host sediments. The variable compositions and morphologies could affect seal integrity. Broader testing of this classification system is recommended to enhance its global applicability.

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